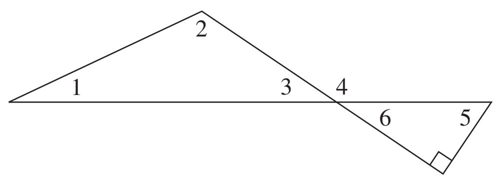
62. If n, m, and p are integers and $n \mid mp$, then $n \mid m$ or $n \mid p$.
63. For every prime number n, $n + 4$ is prime.
64. For every positive integer n, $n^2 + n + 3$ is not prime.
65. For n a positive integer, $n > 2$, $n^2 - 1$ is not prime.
66. For every positive integer n, $2^n + 1$ is prime.
67. For every positive integer n, $n^2 + n + 1$ is prime.
68. For n an even integer, $n > 2$, $2^n - 1$ is not prime.
69. The sum of two rational numbers is rational.
70. The product of two rational numbers is rational.
71. The product of two irrational numbers is irrational.
72. The sum of a rational number and an irrational number is irrational.

For Exercises 73–75, use the following facts from geometry and the accompanying figure.

- The interior angles of a triangle sum to 180°.
- Vertical angles (opposite angles formed when two lines intersect) are the same size.
- A straight angle is 180°.
- A right angle is 90°.



73. Prove that the measure of angle 5 plus the measure of angle 3 is 90°.
74. Prove that the measure of angle 4 is the sum of the measures of angles 1 and 2.
75. If angle 1 and angle 5 are the same size, then angle 2 is a right angle.
76. Prove that the sum of the integers from 1 through 100 is 5050. (*Hint:* Instead of actually adding all the numbers, try to make the same clever observation that the German mathematician Karl Friederick Gauss [1777–1855] made as a schoolchild: Group the numbers into pairs, using 1 and 100, 2 and 99, etc.)

SECTION 2.2 INDUCTION

First Principle of Induction

There is one final proof technique especially useful in computer science. To illustrate how the technique works, imagine that you are climbing an infinitely high ladder. How do you know whether you will be able to reach an arbitrarily high rung? Suppose we make the following two assertions about your climbing abilities:

1. You can reach the first rung.
2. Once you get to a rung, you can always climb to the next one up. (Notice that this assertion is an implication.)

If both statement 1 and the implication of statement 2 are true, then by statement 1 you can get to the first rung and therefore by statement 2 you can get to the second; by statement 2 again, you can get to the third rung; by statement 2 again you can get to the fourth; and so on. You can climb as high as you wish. Both assertions here are necessary. If only statement 1 is true, you have no guarantee of getting beyond the first rung, and if only statement 2 is true, you may never be able to get started. Let's assume that the rungs of the ladder are numbered by positive integers—1, 2, 3, and so on.

Now think of a specific property a number might have. Instead of "reaching an arbitrarily high rung," we can talk about an arbitrary, positive integer having that property. We use the shorthand notation $P(n)$ to mean that the positive integer n has the property P. How can we use the ladder-climbing technique to prove that for all positive integers n, we have $P(n)$? The two assertions we need to prove are

1. $P(1)$	(1 has property P.)
2. For any positive integer k, $P(k) \rightarrow P(k + 1)$.	(If any number has property P, so does the next number.)

If we can prove both assertions 1 and 2, then $P(n)$ holds for any positive integer n, just as you could climb to an arbitrary rung on the ladder.

The foundation for arguments of this type is the first principle of mathematical induction.

PRINCIPLE FIRST PRINCIPLE OF MATHEMATICAL INDUCTION

$$\left.\begin{array}{l} 1.\ P(1) \text{ is true} \\ 2.\ (\forall k)[P(k) \text{ true} \rightarrow P(k+1) \text{ true}] \end{array}\right\} \rightarrow P(n) \text{ true for all positive integers } n$$

REMINDER

To prove something true for all $n \geq$ some value, think induction.

The first principle of mathematical induction is an implication. The conclusion is a statement of the form, "$P(n)$ is true for all positive integers n." Therefore, whenever we want to prove that something is true for every positive integer n, it is a good bet that mathematical induction is an appropriate proof technique to use.

To know that the conclusion of this implication is true, we show that the two hypotheses, statements 1 and 2, are true. To prove statement 1, we need only show that property P holds for the number 1, usually a trivial task. Statement 2 is also an implication that must hold for all k. To prove this implication, we assume for an arbitrary positive integer k that $P(k)$ is true and show, based on this assumption, that $P(k + 1)$ is true. Therefore $P(k) \rightarrow P(k + 1)$ and, using universal generalization, $(\forall k)[P(k) \rightarrow P(k + 1)]$. You should convince yourself that assuming that property P holds for the number k is not the same as assuming what we ultimately want to prove (a frequent source of confusion when one first encounters proofs of this kind). It is merely the way to proceed with a direct proof that the implication $P(k) \rightarrow P(k + 1)$ is true.

In doing a proof by induction, establishing the truth of statement 1, $P(1)$, is called the **basis**, or **basis step**, for the inductive proof. Establishing the truth of $P(k) \rightarrow P(k + 1)$ is called the **inductive step**. When we assume $P(k)$ to be true so as to prove the inductive step, $P(k)$ is called the **inductive assumption**, or **inductive hypothesis**.

All the proof methods we have talked about in this chapter are techniques for deductive reasoning—ways to prove a conjecture that perhaps was formulated by inductive reasoning. Mathematical induction is also a *deductive* technique, not a method for inductive reasoning (don't get confused by the terminology here). For the other proof techniques, we can begin with a hypothesis and string facts together until we more or less stumble on a conclusion. In fact, even if our conjecture is slightly incorrect, we might see what the correct conclusion is in the course of doing the proof. In mathematical induction, however, we must know right at the outset the exact form of the property $P(n)$ that we are trying to establish. Mathematical induction, therefore, is not an exploratory proof technique—it can only confirm a correct conjecture.

Proofs by Mathematical Induction

Suppose that the ancestral progenitor Smith married and had two children. Let's call these two children generation 1. Now suppose each of those two children had two children; then in generation 2, there were four offspring. This trend continued from generation unto generation. The Smith family tree therefore looks like Figure 2.2. (This figure looks exactly like Figure 1.1b, where we looked at the possible T–F values for n statement letters.)

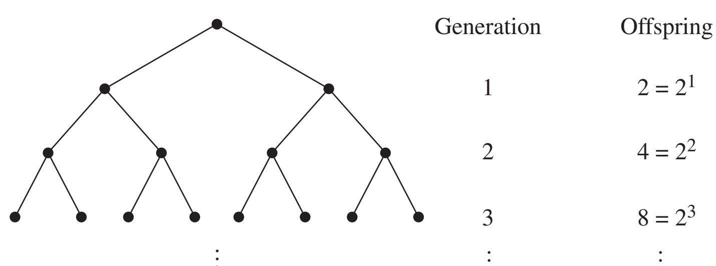


Figure 2.2

It appears that generation n contains 2^n offspring. More formally, if we let $P(n)$ denote the number of offspring at generation n, then we guess that

$$P(n) = 2^n$$

We can use induction to *prove* that our guess for $P(n)$ is correct.

The basis step is to establish $P(1)$, which is the equation

$$P(1) = 2^1 = 2$$

This is true because we are told that Smith had two children. We now assume that our guess is correct for an arbitrary generation k, $k \geq 1$; that is, we assume

$$P(k) = 2^k$$

and try to show that

$$P(k + 1) = 2^{k+1}$$

In this family, each offspring has two children; thus the number of offspring at generation $k + 1$ will be twice the number at generation k, or $P(k + 1) = 2P(k)$. By the inductive assumption, $P(k) = 2^k$, so

$$P(k + 1) = 2P(k) = 2(2^k) = 2^{k+1}$$

and so indeed

$$P(k + 1) = 2^{k+1}$$

This completes our proof. Now that we have set our mind at ease about the Smith clan, we can apply the inductive proof technique to less obvious problems.

EXAMPLE 14 Prove that the equation

$$1 + 3 + 5 + \cdots + (2n - 1) = n^2 \tag{1}$$

is true for any positive integer n. Here the property $P(n)$ is equation (1). (Notice that $P(n)$ is a property of n, or—in language from Chapter 1—a unary predicate. It is a statement about n, expressed here as an equation. Thus it is incorrect to write something like $P(n) = 1 + 3 + 5 + \cdots + (2n - 1)$.)

The left side of this equation is the sum of all the odd integers from 1 to $2n - 1$. The right side is a formula for the value of this sum. Although we can verify the truth of this equation for any particular value of n by substituting that value for n, we cannot substitute all possible positive integer values. Thus a proof by exhaustion does not work. A proof by mathematical induction is appropriate.

The basis step is to establish $P(1)$, which is equation (1) when n has the value 1. When we substitute 1 for n in the left side of equation (1), we get the sum of all the odd integers starting at 1 and ending at $2(1) - 1 = 1$. The sum of all the odd numbers from 1 to 1 equals 1. When we substitute 1 for n in the formula on the right side of this equation, we get $(1)^2$. Therefore

$$P(1): \quad 1 = 1^2$$

This is certainly true. For the inductive hypothesis, we assume $P(k)$ for an arbitrary positive integer k, which is equation (1) when n has the value k, or

$$P(k): \quad 1 + 3 + 5 + \cdots + (2k - 1) = k^2 \tag{2}$$

(Note that $P(k)$ is *not* the equation $(2k - 1) = k^2$, which is true only for $k = 1$.) Using the inductive hypothesis, we want to show $P(k + 1)$, which is equation (1) when n has the value $k + 1$, or

$$P(k + 1): \quad 1 + 3 + 5 + \cdots + [2(k + 1) - 1] \stackrel{?}{=} (k + 1)^2 \tag{3}$$

(The question mark over the equals sign is to remind us that this is the fact we want to prove, as opposed to something we already know.)

The key to an inductive proof is to find a way to relate what we want to show—$P(k+1)$, equation (3)—to what we have assumed—$P(k)$, equation (2). The left side of $P(k+1)$ can be rewritten to show the next-to-last term:

$$1 + 3 + 5 + \cdots + (2k - 1) + [2(k + 1) - 1]$$

This expression contains the left side of equation (2) as a subexpression. Because we have assumed $P(k)$ to be true, we can substitute the right side of equation (2) for this subexpression. Thus,

$$\begin{aligned} &1 + 3 + 5 + \cdots + [2(k + 1) - 1] \\ &\quad = 1 + 3 + 5 + \cdots + (2k - 1) + [2(k + 1) - 1] \\ &\quad = k^2 + [2(k + 1) - 1] \\ &\quad = k^2 + [2k + 2 - 1] \\ &\quad = k^2 + 2k + 1 \\ &\quad = (k + 1)^2 \end{aligned}$$

Therefore,

$$1 + 3 + 5 + \cdots + [2(k + 1) - 1] = (k + 1)^2$$

which verifies $P(k+1)$ and proves that equation (1) is true for any positive integer n. ●

Table 2.3 summarizes the three steps necessary for a proof using the first principle of induction.

TABLE 2.3

To Prove by First Principle of Induction	
Step 1	Prove base case.
Step 2	Assume $P(k)$.
Step 3	Prove $P(k + 1)$.

Any "summation" induction problem works exactly the same way. Write the summation including the next-to-last term, and you will find the left side of the $P(k)$ equation and can use the inductive hypothesis. After that, it's just a question of algebra.

EXAMPLE 15 Prove that

$$1 + 2 + 2^2 + \cdots + 2^n = 2^{n+1} - 1$$

for any $n \geq 1$.

Again, induction is appropriate. $P(1)$ is the equation

$$1 + 2 = 2^{1+1} - 1 \qquad \text{or} \qquad 3 = 2^2 - 1$$

REMINDER

To prove $P(k) \rightarrow P(k+1)$, you have to discover the $P(k)$ case within the $P(k+1)$ case.

which is true. We take $P(k)$

$$1 + 2 + 2^2 + \cdots + 2^k = 2^{k+1} - 1$$

as the inductive hypothesis and try to establish $P(k+1)$:

$$1 + 2 + 2^2 + \cdots + 2^{k+1} \stackrel{?}{=} 2^{k+1+1} - 1$$

Rewriting the sum on the left side of $P(k+1)$ reveals how the inductive assumption can be used:

$$\begin{aligned} &1 + 2 + 2^2 + \cdots + 2^{k+1} \\ &\quad = 1 + 2 + 2^2 + \cdots + 2^k + 2^{k+1} \\ &\quad = 2^{k+1} - 1 + 2^{k+1} && \text{(from the inductive hypothesis P}(k)\text{)} \\ &\quad = 2(2^{k+1}) - 1 && \text{(adding like terms)} \\ &\quad = 2^{k+1+1} - 1 \end{aligned}$$

Therefore,

$$1 + 2 + 2^2 + \cdots + 2^{k+1} = 2^{k+1+1} - 1$$

which verifies $P(k+1)$ and completes the proof. ●

The following summation formula is the most important one because it occurs so often in the analysis of algorithms. If you don't remember anything else, you should remember this formula for the sum of the first n positive integers.

PRACTICE 7 Prove that for any positive integer n,

$$1 + 2 + 3 + \cdots + n = \frac{n(n+1)}{2}$$

■

Not all proofs by induction involve formulas with sums. Other algebraic identities about the positive integers, as well as nonalgebraic assertions like the number of offspring in generation n of the Smith family, can be proved by induction.

EXAMPLE 16 Prove that for any positive integer n, $2^n > n$.

$P(1)$ is the assertion $2^1 > 1$, which is surely true. Now we assume $P(k)$, $2^k > k$, and try to conclude $P(k+1)$, $2^{k+1} > k + 1$. Now, where is $P(k)$ hidden in here? Aha—we can write the left side of $P(k+1)$, 2^{k+1}, as $2k \cdot 2$, and there's the left side of $P(k)$. Using the inductive hypothesis $2^k > k$ and multiplying both sides of this inequality by 2, we get $2^k \cdot 2 > k \cdot 2$. We complete the argument

$$2^{k+1} = 2^k \cdot 2 > k \cdot 2 = k + k \geq k + 1 \text{ (because } k \geq 1)$$

or

$$2^{k+1} > k + 1$$

●

For the first step of the induction process, it may be appropriate to begin at 0 or at 2 or 3 instead of at 1. The same principle applies, no matter where you first hop on the ladder.

EXAMPLE 17 Prove that $n^2 > 3n$ for $n \geq 4$.

Here we should use induction and begin with a basis step of $P(4)$. (Testing values of $n = 1, 2$, and 3 shows that the inequality does not hold for these values.) $P(4)$ is the inequality $4^2 > 3(4)$, or $16 > 12$, which is true. The inductive hypothesis is that $k^2 > 3k$ and that $k \geq 4$, and we want to show that $(k + 1)^2 > 3(k + 1)$.

$$\begin{aligned}
(k+1)^2 &= k^2 + 2k + 1 \\
&> 3k + 2k + 1 && \text{(by the inductive hypothesis)} \\
&\geq 3k + 8 + 1 && \text{(because } k \geq 4\text{)} \\
&= 3k + 9 \\
&> 3k + 3 && \text{(because } 9 > 3\text{)} \\
&= 3(k+1)
\end{aligned}$$

In this proof we used the fact that $3k + 9 > 3k + 3$. Of course, $3k + 9$ is greater than lots of things, but $3k + 3$ is what gives us what we want. In an induction proof, because we know exactly what we want as the result, we can let that guide us as we manipulate algebraic expressions.

PRACTICE 8 Prove that for all $n > 1$,

$$2^{n+1} < 3^n$$

EXAMPLE 18 Prove that for any positive integer n, the number $2^{2n} - 1$ is divisible by 3.

The basis step is to show $P(1)$, that $2^{2(1)} - 1 = 4 - 1 = 3$ is divisible by 3. Clearly this is true.

We assume that $2^{2k} - 1$ is divisible by 3, which means that $2^{2k} - 1 = 3m$ for some integer m, or $2^{2k} = 3m + 1$ (this little rewriting trick is the key to these "divisibility" problems). We want to show that $2^{2(k+1)} - 1$ is divisible by 3.

$$\begin{aligned}
2^{2(k+1)} - 1 &= 2^{2k+2} - 1 \\
&= 2^2 \cdot 2^{2k} - 1 \\
&= 2^2(3m + 1) - 1 && \text{(by the inductive hypothesis)} \\
&= 12m + 4 - 1 \\
&= 12m + 3 \\
&= 3(4m + 1) && \text{where } 4m + 1 \text{ is an integer}
\end{aligned}$$

Thus $2^{2(k+1)} - 1$ is divisible by 3.

Misleading claims of proof by induction are also possible. When we prove the truth of $P(k + 1)$ without relying on the truth of $P(k)$, we have done a direct proof of $P(k + 1)$ where $k + 1$ is arbitrary. The proof is not invalid, but it should be rewritten to show that it is a direct proof of $P(n)$ for any n, not a proof by induction.

An inductive proof may be called for when its application is not as obvious as in the above examples. The problem statement may not directly say "prove something about nonnegative integers." Instead, there is some quantity in the statement to be proved that can take on arbitrary nonnegative integer values.

EXAMPLE 19

A programming language might be designed with the following convention regarding multiplication: A single factor requires no parentheses, but the product "a times b" must be written as $(a)b$. So the product

$$a \cdot b \cdot c \cdot d \cdot e \cdot f \cdot g$$

could be written in this language as

$$((((((a)b)c)d)e)f)g$$

or, for example,

$$((a)b)(((c)d)(e)f)g$$

depending on the order in which the products are formed. The result is the same in either case.

We want to show that any product of factors can be written with an even number of parentheses. The proof is by induction on the number of factors (this is where the nonnegative integer comes in—it represents the number of factors in any product of factors). For a single factor, there are 0 parentheses, an even number. Assume that for any product of k factors there is an even number of parentheses. Now consider a product P of $k + 1$ factors. P can be thought of as r times s where r has k factors and s is a single factor. By the inductive hypothesis, r has an even number of parentheses. Then we write r times s as $(r)s$. This adds 2 more parentheses to the even number of parentheses in r, giving P an even number of parentheses.

Here's an important observation about the proof in Example 19: There are no algebraic expressions! The entire proof is a verbal argument. For some proofs, we can't rely on just the crutch of nonverbal mathematical manipulations; we have to use words.

EXAMPLE 20

A "tiling" problem gives a nice illustration of induction in a geometric setting. An *angle iron* is an L-shaped piece that can cover 3 squares on a checkerboard (Figure 2.3a). The problem is to show that for any positive integer n, a $2^n \times 2^n$ checkerboard with one square removed can be tiled—completely covered—by angle irons.

The base case is $n = 1$, which gives a 2×2 checkerboard. Figure 2.3b shows the solution to this case if the upper right corner is removed. Removing any of the other three corners works the same way. Assume that any $2^k \times 2^k$ checkerboard with one square removed can be tiled using angle irons. Now consider a checkerboard with dimensions $2^{k+1} \times 2^{k+1}$. We need to show that it can be tiled when one square is removed. To relate the $k + 1$ case to the inductive hypothesis, divide the $2^{k+1} \times 2^{k+1}$ checkerboard into four quarters. Each quarter will be a $2^k \times 2^k$ checkerboard, and one will have a missing square (Figure 2.3c). By the inductive hypothesis, this checkerboard can be tiled. Remove a corner from each of the other three checkerboards, as in Figure 2.3d. By the inductive hypothesis, the three boards with the holes removed can be tiled, and one angle iron can tile the three holes. Hence the original $2^{k+1} \times 2^{k+1}$ board with its one hole can be tiled.

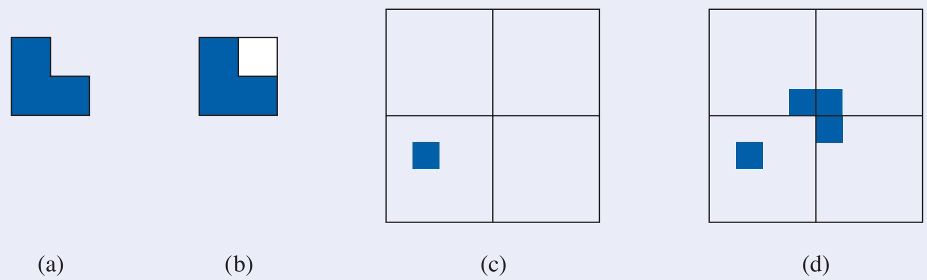


Figure 2.3

Second Principle of Induction

In addition to the first principle of induction, which we have been using,

$$\left.\begin{array}{l} 1.\ \ P(1) \text{ is true} \\ 2.\ \ (\forall k)[P(k) \text{ true} \rightarrow P(k+1) \text{ true}] \end{array}\right\} \rightarrow P(n) \text{ true for all positive integers } n$$

there is a second principle of induction.

PRINCIPLE SECOND PRINCIPLE OF MATHEMATICAL INDUCTION

$$\left.\begin{array}{l} 1'.\ \ P(1) \text{ is true} \\ 2'.\ \ (\forall k)[P(r) \text{ true for all } r, \\ \qquad 1 \le r \le k \rightarrow P(k+1) \text{ true}] \end{array}\right\} \rightarrow P(n) \text{ true for all positive integers } n$$

These two induction principles differ in statements 2 and 2′. In statement 2, we must be able to prove for an arbitrary positive integer k that $P(k + 1)$ is true based only on the assumption that $P(k)$ is true. In statement 2′, we can assume that $P(r)$ is true for all integers r between 1 and an arbitrary positive integer k in order to prove that $P(k + 1)$ is true. This seems to give us a great deal more "ammunition," so we might sometimes be able to prove the implication in 2′ when we cannot prove the implication in 2.

What allows us to deduce $(\forall n)P(n)$ in either case? We will see that the two induction principles themselves, that is, the two methods of proof, are equivalent. In other words, if we accept the first principle of induction as valid, then the second principle of induction is valid, and conversely. In order to prove the equivalence of the two induction principles, we'll introduce another principle, which seems so obvious as to be unarguable.

PRINCIPLE **PRINCIPLE OF WELL-ORDERING**

Every collection of positive integers that contains any members at all has a smallest member.

We will see that the following implications are true:

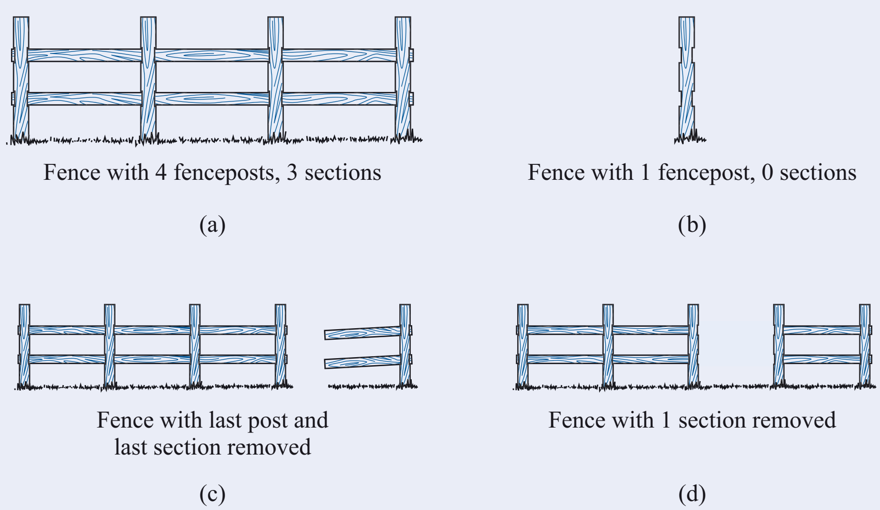
$$\text{second principle of induction} \rightarrow \text{first principle of induction}$$
$$\text{first principle of induction} \rightarrow \text{well-ordering}$$
$$\text{well-ordering} \rightarrow \text{second principle of induction}$$

As a consequence, all three principles are equivalent, and accepting any one of them as true means accepting the other two as well.

To prove that the second principle of induction implies the first principle of induction, suppose we accept the second principle as valid reasoning. We then want to show that the first principle is valid; that is, that we can conclude $P(n)$ for all n from statements 1 and 2. If statement 1 is true, so is statement 1′. If statement 2 is true, then so is statement 2′, because we can say that we concluded $P(k + 1)$ from $P(r)$ for all r between 1 and k, even though we used only the single condition $P(k)$. (More precisely, statement 2′ requires that we prove $P(1) \wedge P(2) \wedge \cdots \wedge P(k) \rightarrow P(k + 1)$, but $P(1) \wedge P(2) \wedge \cdots \wedge P(k) \rightarrow P(k)$, and from statement 2, $P(k) \rightarrow P(k + 1)$, so $P(1) \wedge P(2) \wedge \cdots \wedge P(k) \rightarrow P(k + 1)$.) By the second principle of induction, we conclude $P(n)$ for all n. The proofs that the first principle of induction implies well-ordering and that well-ordering implies the second principle of induction are left as exercises in Section 4.1.

To distinguish between a proof by the first principle of induction and a proof by the second principle of induction, let's look at a rather picturesque example that can be proved both ways.

EXAMPLE 21 Prove that a straight fence with n fence posts has $n - 1$ sections for any $n \geq 1$ (as in Figure 2.4a).



Fence with 4 fenceposts, 3 sections

(a)

Fence with 1 fencepost, 0 sections

(b)

Fence with last post and last section removed

(c)

Fence with 1 section removed

(d)

Figure 2.4

Let $P(n)$ be the statement that a fence with n fence posts has $n - 1$ sections, and prove $P(n)$ true for all $n \geq 1$.

We'll start with the first principle of induction. For the basis step, $P(1)$ says that a fence with only 1 fence post has 0 sections, which is clearly true (Figure 2.4b). Assume that $P(k)$ is true:

a fence with k fence posts has $k - 1$ sections

and try to prove $P(k + 1)$:

(?) a fence with $k + 1$ fence posts has k sections

Given a fence with $k + 1$ fence posts, how can we relate that to a fence with k fence posts so that we can make use of the inductive hypothesis? We can chop off the last post and the last section (Figure 2.4c). The remaining fence has k fence posts and, by the inductive hypothesis, $k - 1$ sections. Therefore the original fence had k sections.

Now we'll prove the same result using the second principle of induction. The basis step is the same as before. For the inductive hypothesis, we assume

for all r, $1 \leq r \leq k$, a fence with r fence posts has $r - 1$ sections

and try to prove $P(k + 1)$:

(?) a fence with $k + 1$ fence posts has k sections

For a fence with $k + 1$ fence posts, split the fence into two parts by removing one section (Figure 2.4d). The 2 parts of the fence have r_1 and r_2 fence posts, where $1 \leq r_1 \leq k$, $1 \leq r_2 \leq k$, and $r_1 + r_2 = k + 1$. By the inductive hypothesis, the 2 parts have, respectively, $r_1 - 1$ and $r_2 - 1$ sections, so the original fence has

$$(r_1 - 1) + (r_2 - 1) + 1 \text{ sections}$$

(The extra 1 is for the one that we removed.) Simple arithmetic then yields

$$r_1 + r_2 - 1 = (k + 1) - 1 = k \text{ sections}$$

This proves that a fence with $k + 1$ fence posts has k sections, which verifies $P(k + 1)$ and completes the proof using the second principle of induction. ●

Example 21 allowed for either form of inductive proof because we could either reduce the fence at one end or split it at an arbitrary point. The problem of Example 19 is similar.

EXAMPLE 22 We again want to show that any product of factors can be written in this programming language with an even number of parentheses, this time using the second principle of induction. The base case is the same as in Example 19: A single factor has 0 parentheses, an even number. Assume that any product of r factors, $1 \leq r \leq k$, can be written with an even number of parentheses. Then consider a product

P with $k + 1$ factors. P can be written as $(S)T$, a product of two factors S and T, where S has r_1 factors and T has r_2 factors. Then $1 \leq r_1 \leq k$ and $1 \leq r_2 \leq k$, with $r_1 + r_2 = k + 1$. By the inductive hypothesis, S and T each have an even number of parentheses, and therefore so does $(S)T = P$.

Most problems do not work equally well with either form of induction; the fence post and the programming language problem were somewhat artificial. Generally, the second principle of induction is called for when the problem "splits" most naturally in the middle instead of growing from the end.

EXAMPLE 23 Prove that for every integer $n \geq 2$, n is a prime number or a product of prime numbers.

We will postpone the decision of whether to use the first or the second principle of induction; the basis step is the same in each case and need not start with 1. Obviously here we should start with 2. $P(2)$ is the statement that 2 is a prime number or a product of primes. Because 2 is a prime number, $P(2)$ is true. Jumping ahead, for either principle we will be considering the number $k + 1$. If $k + 1$ is prime, we are done. If $k + 1$ is not prime, then it is a composite number and can be written as $k + 1 = ab$. Here $k + 1$ has been split into two factors. Maybe neither of these factors has the value k, so an assumption only about $P(k)$ isn't enough. Hence, we'll use the second principle of induction.

So let's start again. We assume that for all r, $2 \leq r \leq k$, $P(r)$ is true—r is prime or the product of primes. Now consider the number $k + 1$. If $k + 1$ is prime, we are done. If $k + 1$ is not prime, then it is a composite number and can be written as $k + 1 = ab$, where $1 < a < k + 1$ and $1 < b < k + 1$. (This is a nontrivial factorization, so neither factor can be 1 or $k + 1$.) Therefore $2 \leq a \leq k$ and $2 \leq b \leq k$. The inductive hypothesis applies to both a and b, so a and b are either prime or the product of primes. Thus, $k + 1 = ab$ is the product of prime numbers. This verifies $P(k + 1)$ and completes the proof by the second principle of induction.

The proof in Example 23 is an *existence* proof rather than a *constructive* proof. Knowing that every nonprime number has a factorization as a product of primes does not make it easy to *find* such a factorization. (We will see in Section 2.4 that there is, except for the order of the factors, only one such factorization.) Some encryption systems for passing information in a secure fashion on the Web depend on the difficulty of factoring large numbers into their prime factors (see the discussion on public-key encryption in Section 5.6).

EXAMPLE 24 Prove that any amount of postage greater than or equal to 8 cents can be built using only 3-cent and 5-cent stamps.

Here we let $P(n)$ be the statement that only 3-cent and 5-cent stamps are needed to build n cents worth of postage, and prove that $P(n)$ is true for all $n \geq 8$. The basis step is to establish $P(8)$, which is done by the equation

$$8 = 3 + 5$$

For reasons that will be clear momentarily, we'll also establish two additional cases, $P(9)$ and $P(10)$, by the equations

$$9 = 3 + 3 + 3$$
$$10 = 5 + 5$$

Now we assume that $P(r)$ is true, that is, r can be written as a sum of 3s and 5s, for any r, $8 \le r \le k$, and consider $P(k + 1)$. We may assume that $k + 1$ is at least 11, because we have already proved $P(r)$ true for $r = 8, 9$, and 10. If $k + 1 \ge 11$, then $(k + 1) - 3 = k - 2 \ge 8$. Thus $k - 2$ is a legitimate r value, and by the inductive hypothesis, $P(k - 2)$ is true. Therefore $k - 2$ can be written as a sum of 3s and 5s, and adding an additional 3 gives us $k + 1$ as a sum of 3s and 5s. This verifies that $P(k + 1)$ is true and completes the proof. ●

PRACTICE 9

a. Why are the additional cases $P(9)$ and $P(10)$ proved separately in Example 24?
b. Why can't the first principle of induction be used in the proof of Example 24? ■

REMINDER

Use the second principle of induction when the $k + 1$ case depends on results farther back than k.

As a general rule, the first principle of mathematical induction applies when information about "one position back" is enough, that is, when the truth of $P(k)$ is enough to prove the truth of $P(k + 1)$. The second principle applies when information about "one position back" isn't good enough; that is, you can't prove that $P(k + 1)$ is true just because you know $P(k)$ is true, but you can prove $P(k + 1)$ true if you know that $P(r)$ is true for one or more values of r that are "farther back" than k.

SECTION 2.2 REVIEW

TECHNIQUES

- Use the first principle of induction in proofs.
- Use the second principle of induction in proofs.

MAIN IDEAS

- Mathematical induction is a technique to prove properties of positive integers.
- An inductive proof need not begin with 1.
- Induction can be used to prove statements about quantities whose values are arbitrary nonnegative integers.
- The first and second principles of induction each prove the same conclusion, but one approach may be easier to use than the other in a given situation.

EXERCISES 2.2

1. For all positive integers, let $P(n)$ be the equation

$$2 + 6 + 10 + \cdots + (4n - 2) = 2n^2$$

a. Write the equation for the base case $P(1)$ and verify that it is true.
b. Write the inductive hypothesis $P(k)$.
c. Write the equation for $P(k + 1)$.
d. Prove that $P(k + 1)$ is true.

2. For all positive integers, let $P(n)$ be the equation

$$2 + 4 + 6 + \cdots + 2n = n(n + 1)$$

a. Write the equation for the base case $P(1)$ and verify that it is true.
b. Write the inductive hypothesis $P(k)$.
c. Write the equation for $P(k + 1)$.
d. Prove that $P(k + 1)$ is true.

In Exercises 3–26, use mathematical induction to prove that the statements are true for every positive integer n. [*Hint*: In the algebra part of the proof, if the final expression you want has factors and you can pull those factors out early, do that instead of multiplying everything out and getting some humongous expression.]

3. $1 + 5 + 9 + \cdots + (4n - 3) = n(2n - 1)$
4. $1 + 3 + 6 + \cdots + \dfrac{n(n + 1)}{2} = \dfrac{n(n + 1)(n + 2)}{6}$
5. $4 + 10 + 16 + \cdots + (6n - 2) = n(3n + 1)$
6. $5 + 10 + 15 + \cdots + 5n = \dfrac{5n(n + 1)}{2}$
7. $1^2 + 2^2 + \cdots + n^2 = \dfrac{n(n + 1)(2n + 1)}{6}$
8. $1^3 + 2^3 + \cdots + n^3 = \dfrac{n^2(n + 1)^2}{4}$
9. $1^2 + 3^2 + \cdots + (2n - 1)^2 = \dfrac{n(2n - 1)(2n + 1)}{3}$
10. $1^4 + 2^4 + \cdots + n^4 = \dfrac{n(n + 1)(2n + 1)(3n^2 + 3n - 1)}{30}$
11. $1 \cdot 3 + 2 \cdot 4 + 3 \cdot 5 + \cdots + n(n + 2) = \dfrac{n(n + 1)(2n + 7)}{6}$
12. $1 + a + a^2 + \cdots + a^{n-1} = \dfrac{a^n - 1}{a - 1}$ for $a \neq 0, a \neq 1$
13. $\dfrac{1}{1 \cdot 2} + \dfrac{1}{2 \cdot 3} + \dfrac{1}{3 \cdot 4} + \cdots + \dfrac{1}{n(n + 1)} = \dfrac{n}{n + 1}$
14. $\dfrac{1}{1 \cdot 3} + \dfrac{1}{3 \cdot 5} + \dfrac{1}{5 \cdot 7} + \cdots + \dfrac{1}{(2n - 1)(2n + 1)} = \dfrac{n}{2n + 1}$
15. $1^2 - 2^2 + 3^2 - 4^2 + \cdots + (-1)^{n+1}n^2 = \dfrac{(-1)^{n+1}(n)(n + 1)}{2}$
16. $2 + 6 + 18 + \cdots + 2 \cdot 3^{n-1} = 3^n - 1$
17. $2^2 + 4^2 + \cdots + (2n)^2 = \dfrac{2n(n + 1)(2n + 1)}{3}$
18. $1 \cdot 2^1 + 2 \cdot 2^2 + 3 \cdot 2^3 + \cdots + n \cdot 2^n = (n - 1)2^{n+1} + 2$
19. $1 \cdot 2 + 2 \cdot 3 + 3 \cdot 4 + \cdots + n(n + 1) = \dfrac{n(n + 1)(n + 2)}{3}$

20. $1 \cdot 2 \cdot 3 + 2 \cdot 3 \cdot 4 + \cdots + n(n + 1)(n + 2) = \dfrac{n(n + 1)(n + 2)(n + 3)}{4}$

21. $\dfrac{1}{1 \cdot 4} + \dfrac{1}{4 \cdot 7} + \dfrac{1}{7 \cdot 10} + \cdots + \dfrac{1}{(3n - 2)(3n + 1)} = \dfrac{n}{3n + 1}$

22. $1 \cdot 1! + 2 \cdot 2! + 3 \cdot 3! + \cdots + n \cdot n! = (n + 1)! - 1$ where $n!$ is the product of the positive integers from 1 to n.

23. $1 + 4 + 4^2 + \cdots + 4^n = \dfrac{4^{n+1} - 1}{3}$

24. $1 + x + x^2 + \cdots + x^n = \dfrac{x^{n+1} - 1}{x - 1}$ where x is any integer > 1

25. $1 + 4 + 7 + 10 + \cdots + (3n - 2) = \dfrac{n(3n - 1)}{2}$

26. $1 + [x \cdot 2 - (x - 1)] + [x \cdot 3 - (x - 1)] + \cdots + [x \cdot n - (x - 1)] = \dfrac{n[xn - (x - 2)]}{2}$ where x is any integer ≥ 1

27. A *geometric progression* (*geometric sequence*) is a sequence of terms where there is an initial term a and each succeeding term is obtained by multiplying the previous term by a *common ratio* r. Prove the formula for the sum of the first n $(n \geq 1)$ terms of a geometric sequence where $r \neq 1$:

$$a + ar + ar^2 + \cdots + ar^{n-1} = \frac{a - ar^n}{1 - r}$$

28. An *arithmetic progression* (*arithmetic sequence*) is a sequence of terms where there is an initial term a and each succeeding term is obtained by adding a *common difference* d to the previous term. Prove the formula for the sum of the first n $(n \geq 1)$ terms of an arithmetic sequence:

$$a + (a + d) + (a + 2d) + \cdots + [a + (n - 1)d] = \frac{n}{2}[2a + (n - 1)d]$$

29. Using Exercises 27 and 28, find an expression for the value of the following sums.
   a. $2 + 2 \cdot 5 + 2 \cdot 5^2 + \cdots + 2 \cdot 5^9$
   b. $4 \cdot 7 + 4 \cdot 7^2 + 4 \cdot 7^3 + \cdots + 4 \cdot 7^{12}$
   c. $1 + 7 + 13 + \cdots + 49$
   d. $12 + 17 + 22 + 27 + \cdots + 92$

30. Prove that

$$(-2)^0 + (-2)^1 + (-2)^2 + \cdots + (-2)^n = \frac{1 - 2^{n+1}}{3}$$

for every positive odd integer n.

31. Prove that $n^2 > n + 1$ for $n \geq 2$.
32. Prove that $n^2 \geq 2n + 3$ for $n \geq 3$.
33. Prove that $n^2 > 5n + 10$ for $n > 6$.
34. Prove that $2^n > n^2$ for $n \geq 5$.

In Exercises 35–40, $n!$ is the product of the positive integers from 1 to n.

35. Prove that $n! > n^2$ for $n \geq 4$.

36. Prove that $n! > n^3$ for $n \geq 6$.
37. Prove that $n! > 2^n$ for $n \geq 4$.
38. Prove that $n! > 3^n$ for $n \geq 7$.
39. Prove that $n! \geq 2^{n-1}$ for $n \geq 1$.
40. Prove that $n! < n^n$ for $n \geq 2$.
41. Prove that $(1 + x)^n > 1 + x^n$ for $n > 1, x > 0$.
42. Prove that $\left(\dfrac{a}{b}\right)^{n+1} < \left(\dfrac{a}{b}\right)^n$ for $n \geq 1$ and $0 < a < b$.
43. Prove that $1 + 2 + \cdots + n < n^2$ for $n > 1$.
44. Prove that $1 + \dfrac{1}{4} + \dfrac{1}{9} + \cdots + \dfrac{1}{n^2} < 2 - \dfrac{1}{n} \qquad \text{for } n \geq 2$
45. a. Try to use induction to prove that

$$1 + \frac{1}{2} + \frac{1}{4} + \cdots + \frac{1}{2^n} < 2 \qquad \text{for } n \geq 1$$

What goes wrong?

b. Prove that

$$1 + \frac{1}{2} + \frac{1}{4} + \cdots + \frac{1}{2^n} = 2 - \frac{1}{2^n} \qquad \text{for } n \geq 1$$

thus showing that

$$1 + \frac{1}{2} + \frac{1}{4} + \cdots + \frac{1}{2^n} < 2 \qquad \text{for } n \geq 1$$

46. Prove that

$$1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{2^n} \geq 1 + \frac{n}{2} \qquad \text{for } n \geq 1$$

(Note that the denominators increase by 1, not by powers of 2.)

For Exercises 47–58, prove that the statements are true for every positive integer.

47. $2^{3n} - 1$ is divisible by 7.
48. $3^{2n} + 7$ is divisible by 8.
49. $7^n - 2^n$ is divisible by 5.
50. $13^n - 6^n$ is divisible by 7.
51. $2^n + (-1)^{n+1}$ is divisible by 3.
52. $2^{5n+1} + 5^{n+2}$ is divisible by 27.
53. $3^{4n+2} + 5^{2n+1}$ is divisible by 14.
54. $7^{2n} + 16n - 1$ is divisible by 64.
55. $10^n + 3 \cdot 4^{n+2} + 5$ is divisible by 9.
56. $n^3 - n$ is divisible by 3.
57. $n^3 + 2n$ is divisible by 3.
58. $x^n - 1$ is divisible by $x - 1$ for $x \neq 1$.

59. Prove DeMoivre's Theorem:

$$(\cos \theta + i \sin \theta)^n = \cos n\theta + i \sin n\theta$$

for all $n \geq 1$. *Hint:* Recall the addition formulas from trigonometry:

$$\cos(\alpha + \beta) = \cos \alpha \cos \beta - \sin \alpha \sin \beta$$
$$\sin(\alpha + \beta) = \sin \alpha \cos \beta + \cos \alpha \sin \beta$$

60. Prove that

$$\sin \theta + \sin 3\theta + \cdots + \sin(2n - 1)\theta = \frac{\sin^2 n\theta}{\sin \theta}$$

for all $n \geq 1$ and all θ for which $\sin \theta \neq 0$.

61. Use induction to prove that the product of any three consecutive positive integers is divisible by 3.

62. Suppose that exponentiation is defined by the equation

$$x^j \cdot x = x^{j+1}$$

for any $j \geq 1$. Use induction to prove that $x^n \cdot x^m = x^{n+m}$ for $n \geq 1$, $m \geq 1$.
(*Hint:* Do induction on m for a fixed, arbitrary value of n.)

63. According to Example 20, it is possible to use angle irons to tile a 4×4 checkerboard with the upper right corner removed.Sketch such a tiling.

64. Example 20 does not cover the case of checkerboards that are not sized by powers of 2. Determine whether it is possible to tile a 3×3 checkerboard.

65. Prove that it is possible to use angle irons to tile a 5×5 checkerboard with the upper left corner removed.

66. Find a configuration for a 5×5 checkerboard with one square removed that is not possible to tile; explain why this is not possible.

67. Consider n infinitely long straight lines, none of which are parallel and no three of which have a common point of intersection. Show that for $n \geq 1$, the lines divide the plane into $(n^2 + n + 2)/2$ separate regions.

68. A string of 0*s* and 1*s* is to be processed and converted to an even-parity string by adding a parity bit to the end of the string.(For an explanation of the use of parity bits, see Example 30 in Chapter 9.) The parity bit is initially 0. When a 0 character is processed, the parity bit remains unchanged. When a 1 character is processed, the parity bit is switched from 0 to 1 or from 1 to 0. Prove that the number of 1*s* in the final string, that is, including the parity bit, is always even. (*Hint:* Consider various cases.)

69. What is wrong with the following "proof" by mathematical induction? We will prove that for any positive integer n, n is equal to 1 more than n. Assume that $P(k)$ is true.

$$k = k + 1$$

Adding 1 to both sides of this equation, we get

$$k + 1 = k + 2$$

Thus,

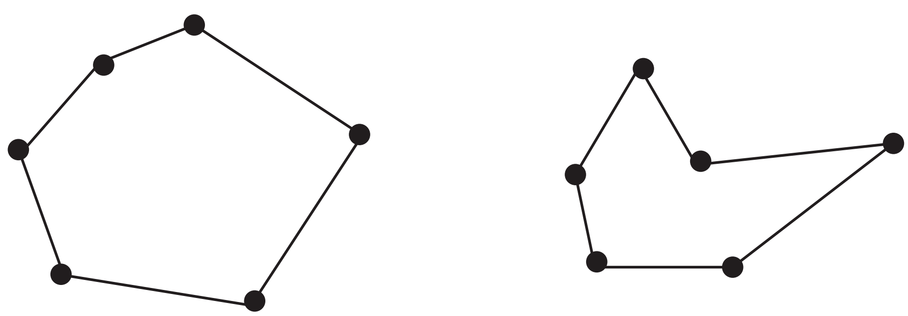
$$P(k + 1) \text{ is true}$$

70. What is wrong with the following "proof" by mathematical induction?

> We will prove that all computers are built by the same manufacturer. In particular, we will prove that in any collection of n computers where n is a positive integer, all the computers are built by the same manufacturer. We first prove $P(1)$, a trivial process, because in any collection consisting of

one computer, there is only one manufacturer. Now we assume $P(k)$; that is, in any collection of k computers, all the computers were built by the same manufacturer. To prove $P(k + 1)$, we consider any collection of $k + 1$ computers. Pull one of these $k + 1$ computers (call it HAL) out of the collection. By our assumption, the remaining k computers all have the same manufacturer. Let HAL change places with one of these k computers. In the new group of k computers, all have the same manufacturer. Thus, HAL's manufacturer is the same one that produced all the other computers, and all $k + 1$ computers have the same manufacturer.

71. An obscure tribe has only three words in its language, *moon*, *noon*, and *soon*. New words are composed by juxtaposing these words in any order, as in *soonnoonmoonnoon*. Any such juxtaposition is a legal word.
    a. Use the first principle of induction (on the number of subwords in the word) to prove that any word in this language has an even number of o's.
    b. Use the second principle of induction (on the number of subwords in the word) to prove that any word in this language has an even number of o's.
72. A *simple closed polygon* consists of n points in the plane joined in pairs by n line segments; each point is the endpoint of exactly 2 line segments. Following are two examples.



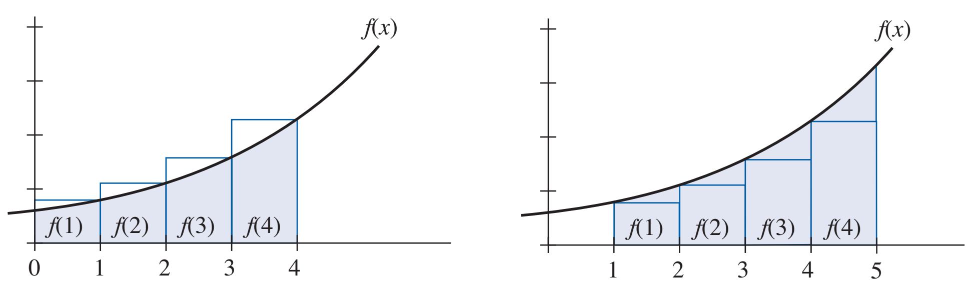
    a. Use the first principle of induction to prove that the sum of the interior angles of an n-sided simple closed polygon is $(n - 2)180°$ for all $n \geq 3$.
    b. Use the second principle of induction to prove that the sum of the interior angles of an n-sided simple closed polygon is $(n - 2)180°$ for all $n \geq 3$.
73. The Computer Science club is sponsoring a jigsaw puzzle contest. Jigsaw puzzles are assembled by fitting 2 pieces together to form a small block, adding a single piece to a block to form a bigger block, or fitting 2 blocks together. Each of these moves is considered a step in the solution. Use the second principle of induction to prove that the number of steps required to assemble an n-piece jigsaw puzzle is $n - 1$.
74. OurWay Pizza makes only two kinds of pizza, pepperoni and vegetarian. Any pizza of either kind comes with an even number of breadsticks (not necessarily the same even number for both kinds). Any order of 2 or more pizzas must include at least 1 of each kind. When the delivery driver goes to deliver an order, he or she puts the completed order together by combining 2 suborders—picking up all the pepperoni pizzas from 1 window and all the vegetarian pizzas from another window. Prove that for a delivery of n pizzas, $n \geq 1$, there are an even number of breadsticks included.
75. Consider propositional wffs that contain only the connectives $\wedge$, $\vee$, and $\rightarrow$ (no negation) and where wffs must be parenthesized when joined by a logical connective. Count each statement letter, connective, or parenthesis as one symbol. For example, $((A) \wedge (B)) \vee ((C) \wedge (D))$ is such a wff, with 19 symbols. Prove that any such wff has an odd number of symbols.
76. In any group of k people, $k \geq 1$, each person is to shake hands with every other person. Find a formula for the number of handshakes, and prove the formula using induction.

77. Prove that any amount of postage greater than or equal to 2 cents can be built using only 2-cent and 3-cent stamps.
78. Prove that any amount of postage greater than or equal to 12 cents can be built using only 4-cent and 5-cent stamps.
79. Prove that any amount of postage greater than or equal to 14 cents can be built using only 3-cent and 8-cent stamps.
80. Prove that any amount of postage greater than or equal to 42 cents can be built using only 4-cent and 15-cent stamps.
81. Prove that any amount of postage greater than or equal to 64 cents can be built using only 5-cent and 17-cent stamps.
82. Your bank ATM delivers cash using only \$20 and \$50 bills. Prove that you can collect, in addition to \$20, any multiple of \$10 that is \$40 or greater.

Exercises 83–84 require familiarity with ideas from calculus. Exercises 1–26 give exact formulas for the sum of terms in a sequence that can be expressed as $\sum_{m=1}^{n} f(m)$. Sometimes it is difficult to find an exact expression for this summation, but if the value of $f(m)$ increases monotonically, integration can be used to find upper and lower bounds on the value of the summation. Specifically,

$$\int_0^n f(x)dx \le \sum_{m=1}^{n} f(m) \le \int_1^{n+1} f(x)dx$$

Using the following figure, we can see (on the left) that $\int_0^n f(x)dx$ underestimates the value of the summation while (on the right) $\int_1^{n+1} f(x)dx$ overestimates it.



83. Show that $\int_0^n 2x\,dx \le \sum_{m=1}^{n} 2m \le \int_1^{n+1} 2x\,dx$ (see Exercise 2).

84. Show that $\int_0^n x^2dx \le \sum_{m=1}^{n} m^2 \le \int_1^{n+1} x^2dx$ (see Exercise 7).